

Project Overview

Project name : IT Asset Management System

Purpose:

A system for tracking and managing an organization's IT assets, including their assignments, locations, statuses, requests, and maintenance history

1. Admin
2. IT Technician
3. Employee
4. Manager / Viewer

Core entities:

Asset

Employee

Branch

User

Asset Request

Assignment

Maintenance Record

Damage Incident

Audit Log

Compensation Decision

Role

Permission

Functional Requirements

FR-01: The system shall allow authorized users to register new IT assets.

FR-02: The system shall allow authorized users to update asset information.

FR-03: The system shall allow authorized users to view, search, and filter registered assets.

FR-04: The system shall allow IT technicians to assign an available asset directly to an employee or in response to an approved asset request.

FR-05: The system shall allow IT technicians to record the return of an assigned asset.

FR-06: The system shall allow IT technicians to record maintenance activities for an asset.

FR-07: The system shall track the current status of each asset.

FR-08: The system shall maintain asset assignment and maintenance history.

FR-09: The system shall allow administrators to manage users and branches.

FR-10: The system shall restrict system functions according to user roles and permissions.

FR-11: The system shall allow employees to submit requests for IT assets.

FR-12: The system shall allow employees to request the return or replacement of assigned assets.

FR-13: The system shall allow IT technicians to review, approve, or reject asset requests.

FR-14: The system shall allow employees to view their assigned assets and track the status of their requests.

FR-15: The system shall notify employees when their request status changes and provide pickup instructions when an assigned asset is ready for collection.

FR-16: The system shall allow managers to view reports and summaries about assets, requests, assignments, and maintenance.

FR-17: The system shall allow IT technicians to record asset damage incidents, including damage details, reported employee responsibility, and estimated repair cost.

FR-18: The system shall allow administrators to record and track authorized compensation or deduction decisions related to employee-caused asset damage.

FR-19: The system shall allow registered users to authenticate using their credentials.

FR-20: The system shall allow registered users to logout directly.

Non-Functional Requirements

NFR-01: The system should have a simple and clear interface.

NFR-02: The system should validate all required input.

NFR-03: The system should protect restricted operations.

NFR-04: The code should follow object-oriented principles.

NFR-05: The system should use appropriate design patterns.

NFR-06: The project should include tests and documentation.

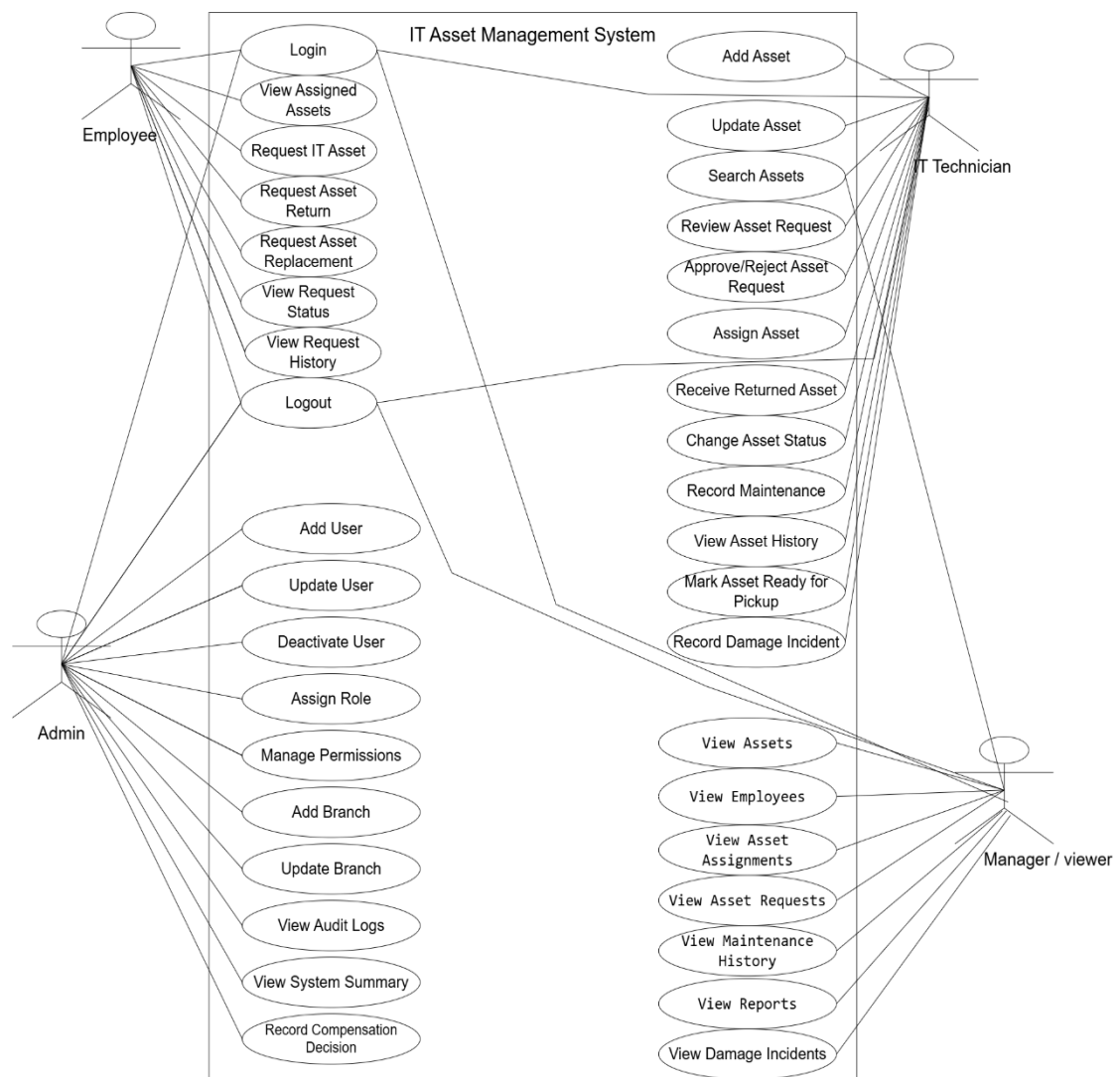
NFR-07: The system should securely store user passwords using hashing.

NFR-08: The system should enforce authorization based on user roles and permissions.

NFR-09: The system should record important system actions for auditing and traceability.

NFR-10: The system should provide acceptable response times for common user operations.

Use Case Analysis



Actor: Admin

Description: An administrator who manages users, roles, branches, and access permissions.

Use Case	Description
Login	The administrator signs into the system.
Add User	The administrator creates a new system user account.
Update User	The administrator updates user information.
Deactivate User	The administrator disables a user account without deleting its history.
Assign Role	The administrator assigns a role to a user.
Manage Permissions	The administrator controls what each role can access or modify.
Add Branch	The administrator registers a new company branch.
Update Branch	The administrator updates branch information.
View Audit Logs	The administrator reviews important actions performed in the system.
View System Summary	The administrator views a general summary of users, branches, assets, and requests.
Record Compensation Decision	The administrator records an authorized compensation or deduction decision related to an asset damage incident.
Logout	The administrator signs out of the system.

Actor: IT Technician

Description: An IT technician who manages IT assets, approves requests, assigns devices, performs maintenance, and updates asset information.

Use Case	Description
Login	The IT technician signs into the system.
Add Asset	The IT technician registers a new IT asset in the system.
Update Asset	The IT technician updates asset information such as name, type, serial number, or status.
Search Assets	The IT technician searches for assets using different criteria.
Review Asset Request	The IT technician reviews an employee's asset request before making a decision.
Approve/Reject Asset Request	The IT technician approves or rejects a reviewed asset request.
Assign Asset	The IT technician assigns an available asset directly to an employee or to an approved asset request.
Receive Returned Asset	The IT technician receives a returned asset from an employee.
Change Asset Status	The IT technician changes the asset status to Available, Assigned, Under Maintenance, Retired, or Lost.
Record Maintenance	The IT technician records maintenance details for an asset.
View Asset History	The IT technician views assignment and maintenance history of an asset.
Mark Asset Ready for Pickup	The IT technician marks an assigned asset as ready for collection and specifies the pickup location or instructions.
Record Damage Incident	The IT technician records asset damage details, the potentially responsible employee, and the estimated repair cost.
Logout	The IT technician signs out of the system.

Actor: Employee

Description: An employee who requests and views IT assets, submits return or replacement requests, and tracks asset requests through the system.

Use Case	Description
Login	The employee signs into the system.
View Assigned Assets	The employee views the IT assets currently assigned to them.
Request IT Asset	The employee submits a request for a new IT asset.
Request Asset Return	The employee requests to return an assigned asset.
Request Asset Replacement	The employee requests a replacement for an assigned asset.
View Request Status	The employee tracks the status of submitted requests and views pickup instructions when an asset is ready for collection.
View Request History	The employee views previous asset requests.
Logout	The employee signs out of the system.

Actor: Manager / Viewer

Description: A manager/viewer who monitors IT assets, employees, requests, maintenance records, damage incidents, and reports without modifying system data.

Use Case	Description
Login	The manager signs into the system.
View Assets	The manager views all registered IT assets.
Search Assets	The manager searches for assets.
View Employees	The manager views employee information.
View Asset Assignments	The manager reviews asset assignments.
View Asset Requests	The manager reviews employee asset requests.
View Maintenance History	The manager views maintenance records.
View Reports	The manager views system reports and summaries.
View Damage Incidents	The manager views recorded asset damage incidents and related costs.
Logout	The manager signs out of the system.

Business Rules

Rule ID	Business Rule
BR-01	Only assets with the status Available can be assigned.
BR-02	An asset can be assigned to only one employee at a time.
BR-03	An employee cannot have two pending requests for the same asset type.
BR-04	An asset may be assigned directly by an IT technician or through an approved asset request
BR-05	Returned assets must be inspected before becoming available again.
BR-06	Every maintenance operation must be recorded in the asset history.
BR-07	Users can perform only the actions permitted by their assigned roles.
BR-08	Employees can view only their own assigned assets and requests.
BR-09	Users with existing history must be deactivated, and assets with existing history must be retired rather than permanently deleted.
BR-10	Every asset damage incident must be recorded before any compensation or deduction decision is tracked.
BR-11	IT technicians may record damage details but cannot authorize financial deductions.
BR-12	Pickup instructions shall be provided only after an asset has been assigned and is ready for .collection
BR-13	Important administrative and asset operations must be recorded in the audit log.

System Status Definitions

Asset Request Status:

PENDING - Request submitted and waiting for review.

UNDER_REVIEW - Request is being reviewed by IT.

APPROVED - Request has been approved.

REJECTED - Request has been rejected.

IN_PROGRESS - The approved request is currently being processed.

COMPLETED - The requested action has been completed.

Asset Status:

AVAILABLE - Asset is available for assignment.

ASSIGNED - Asset is currently assigned to an employee.

UNDER_MAINTENANCE - Asset is undergoing maintenance.

RETIRED - Asset has been removed from active service.

LOST - Asset has been reported lost.

Account Status:

ACTIVE - The user account is active and can access the system.

INACTIVE - The user account is deactivated and cannot access the system.

Design Notes

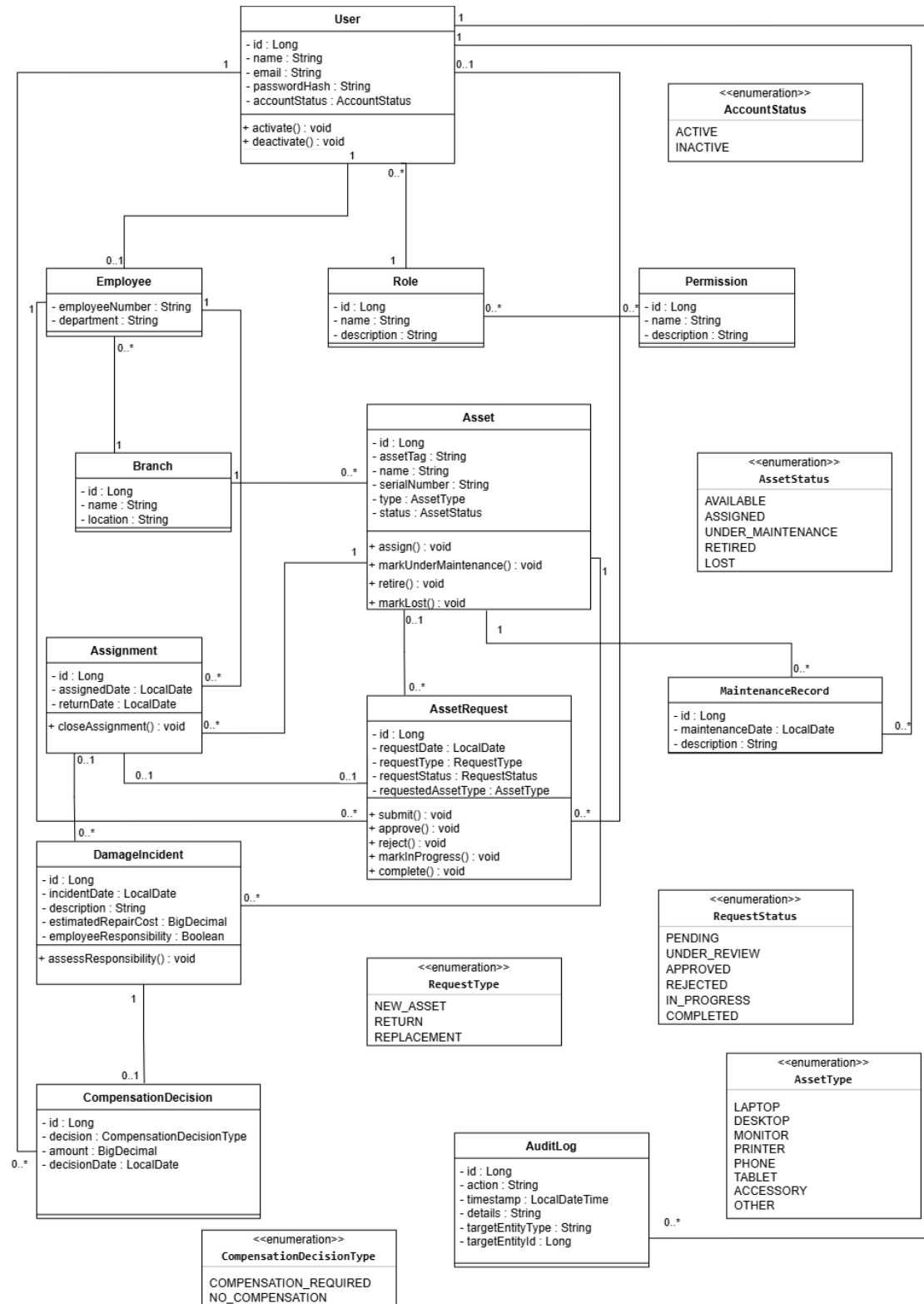
Notification Handling: Notifications will be handled by a notification service and will not be stored as a separate entity in the current system scope.

Asset Assignment: Assets may be assigned directly by an IT Technician or through an approved asset request.

Historical Data: Users and assets with existing historical records will not be permanently deleted. Users will be deactivated and assets will be retired.

Authorization: System operations will be controlled based on user roles and permissions.

Class Diagram Analysis



Candidate Classes

1. User
2. Employee
3. Asset
4. Branch
5. AssetRequest
6. Assignment
7. MaintenanceRecord
8. DamageIncident
9. AuditLog
10. CompensationDecision
11. Role
12. Permission

User

Attributes:

- id : Long
- name : String
- email : String
- passwordHash : String
- accountStatus : AccountStatus

Operations:

- + activate() : void
- + deactivate() : void

AccountStatus:

- ACTIVE
- INACTIVE

Employee

Attributes:

- employeeNumber : String
- department : String

Role

Attributes:

- id : Long
- name : String
- description : String

Permission

Attributes:

- id : Long
- name : String
- description : String

Asset

Attributes:

- id : Long
- assetTag : String
- name : String
- serialNumber : String
- type : AssetType
- status : AssetStatus

Operations:

- + assign() : void
- + markUnderMaintenance() : void
- + retire() : void
- + markLost() : void

AssetStatus:

- AVAILABLE
- ASSIGNED
- UNDER_MAINTENANCE
- RETIRED
- LOST

AssetType:

- LAPTOP
- DESKTOP
- MONITOR
- PRINTER
- PHONE
- TABLET
- ACCESSORY
- OTHER

Branch

Attributes:

- id : Long
- name : String
- location : String

AssetRequest

Attributes:

- id : Long
- requestDate : LocalDate
- requestType : RequestType
- requestStatus : RequestStatus
- requestedAssetType : AssetType

Operations:

- + submit() : void
- + approve() : void
- + reject() : void
- + markInProgress() : void
- + complete() : void

RequestType:

- NEW_ASSET
- RETURN
- REPLACEMENT

RequestStatus:

- PENDING
- UNDER_REVIEW
- APPROVED
- REJECTED
- IN_PROGRESS
- COMPLETED

Assignment

Attributes:

- id : Long
- assignedDate : LocalDate
- returnDate : LocalDate

Operations:

- + closeAssignment() : void

MaintenanceRecord

Attributes:

- id : Long
- maintenanceDate : LocalDate
- description : String

DamageIncident

Attributes:

- id : Long
- incidentDate : LocalDate
- description : String
- estimatedRepairCost : BigDecimal
- employeeResponsibility : Boolean

Operations:

- + assessResponsibility() : void

CompensationDecision

Attributes:

- id : Long
- decision : CompensationDecisionType
- amount : BigDecimal
- decisionDate : LocalDate

CompensationDecisionType:

- COMPENSATION_REQUIRED
- NO_COMPENSATION

AuditLog

Attributes:

- id : Long
- action : String
- timestamp : LocalDateTime
- details : String
- targetEntityType : String
- targetEntityId : Long

Identified Relationships

1. User - Employee

Relationship Type: Association

Multiplicity: User 1 ----- 0..1 Employee

A user account may be associated with zero or one employee profile, while each employee profile belongs to exactly one user account.

2. Branch - Employee

Relationship Type: Association

Multiplicity: Branch 1 ----- 0..* Employee

Each employee belongs to one branch, while a branch may contain many employees.

3. Branch - Asset

Relationship Type: Association

Multiplicity: Branch 1 ----- 0..* Asset

Each asset belongs to one branch, while a branch may contain many assets.

4.Employee - AssetRequest

Relationship Type: Association

Multiplicity: Employee 1 ----- 0..* AssetRequest

An employee may submit multiple asset requests, while each request belongs to one employee.

5.Employee - Assignment

Relationship Type: Association

Multiplicity: Employee 1 ----- 0..* Assignment

An employee may have multiple assignments over time, while each assignment belongs to one employee.

6.Asset - Assignment

Relationship Type: Association

Multiplicity: Asset 1 ----- 0..* Assignment

An asset may have multiple assignment records over time, while each assignment refers to one asset.

Constraint: An asset can have only one active assignment at a time.

7.AssetRequest - Assignment

Relationship Type: Association

Multiplicity: AssetRequest 0..1 ----- 0..1 Assignment

An approved asset request may result in one assignment. An assignment may also be created directly without an asset request.

8.Asset - MaintenanceRecord

Relationship Type: Association

Multiplicity: Asset 1 ----- 0..* MaintenanceRecord

An asset may have multiple maintenance records, while each maintenance record belongs to one asset.

9.User - MaintenanceRecord

Relationship Type: Association

Multiplicity: User 1 ----- 0..* MaintenanceRecord

A user acting as an IT Technician may record multiple maintenance operations, while each maintenance record is recorded by one user.

10.Asset - DamageIncident

Relationship Type: Association

Multiplicity: Asset 1 ----- 0..* DamageIncident

An asset may have multiple damage incidents, while each damage incident refers to one asset.

11.Assignment - DamageIncident

Relationship Type: Association

Multiplicity: Assignment 0..1 ----- 0..* DamageIncident

An assignment may be associated with multiple damage incidents, while each damage incident may be associated with zero or one assignment.

12.DamageIncident - CompensationDecision

Relationship Type: Association

Multiplicity: DamageIncident 1 ----- 0..1 CompensationDecision

A damage incident may result in zero or one compensation decision, while each compensation decision belongs to one damage incident.

13.User - CompensationDecision

Relationship Type: Association

Multiplicity: User 1 ----- 0..* CompensationDecision

A user acting as an Admin may record multiple compensation decisions, while each compensation decision is recorded by one user.

14.User - AuditLog

Relationship Type: Association

Multiplicity: User 1 ----- 0..* AuditLog

A user may generate multiple audit log entries, while each audit log entry records an action performed by one user.

15.User - Role

Relationship Type: Association

Multiplicity: Role 1 ----- 0..* User

Each user is assigned one role, while a role may be assigned to multiple users.

16.Role - Permission

Relationship Type: Association

Multiplicity: Role 0..* ----- 0..* Permission

A role may contain multiple permissions, and a permission may be assigned to multiple roles.

17.Asset - AssetRequest

Relationship Type: Association

Multiplicity: Asset 0..1 ----- 0..* AssetRequest

A request may reference one existing asset when the request type is RETURN or REPLACEMENT, while an asset may be referenced by multiple requests over time.

18.User - AssetRequest

Relationship Type: Association

Multiplicity: User 0..* ----- 0..1 AssetRequest

Each AssetRequest may be reviewed by zero or one User acting as an IT Technician, while a User may review multiple AssetRequests.

Document History

Version	Date	Description
1.0	2026-08-06	Initial project overview and requirements
1.1	2026-08-06	Added actor use case analysis
1.2	2026-08-06	Added business rules
1.3	2026-08-07	Added direct asset assignment, damage incident tracking, compensation records, pickup notifications, and system status definitions.
1.4	2026-08-09	Added Use Case Diagram
1.5	2026-08-09	Added Class Diagram Analysis and identified relationships
1.6	2026-08-10	Updated class analysis, added dynamic role and permission management, authentication requirements, asset request details, and damage incident .attributes
1.7	2026-08-10	Updated request status definitions, refined the User–Employee relationship, and added the Asset–AssetRequest relationship.
1.8	2026-08-10	Added request reviewer tracking and enhanced audit logging with target entity information.
1.9	2026-08-13	Finalized Class Diagram attributes, operations, enumerations, and relationship multiplicities.
1.10	2026-08-15	Finalized requirements, actor descriptions, business rules, system status definitions, design notes, and class diagram documentation.